

Solution

SNP-03

Class 08 - Science

1. The device which is used to protect the buildings from the effect of lightning.
2. The process of transferring of charge from a charged body to the earth is called earthing.
3. Earthquake is caused by a disturbance deep inside the earth's crust.
4. During thunderstorms avoid going at an open place, try to rush towards safe place like in a house or building, if you are travelling in nay vehicle like car or bus, you are safe inside with windows and doors of the vehicle closed. If you are in a forest take shelter under a smaller tree.
5. The process of transferring of charge from a charged object to the earth is called earthing. For our safety, most of the electrical appliances and the mains of the house are connected to earth, so that we can be prevented from getting an electric shock. Earthing is provided in building to protect us from electrical shocks due to any leakage of electrical current.
6. The infamous tsunami in the Indian Ocean which came on 26th December 2004 was because of a massive earthquake near Java.
7. No. It is not true.
8. Lightning rod conductor is a device used to protect buildings from the effects of lightning.
9. During thunderstorm avoid going at an open place, try to rush towards safe place like in a house or building, if you are travelling in nay vehicle like car or bus, you are safe inside with windows and doors of the vehicle closed. If you are in a forest take shelter under a smaller tree.
10. On the top of the building.
11. Tremors on earth can also be caused when a volcano erupts, underground nuclear explosion and meteor hits the earth.
12. No, it is not safe.
13. Thunderstorm, lightning and cyclones.
14. Vibrations caused by an earthquake travel in the form of waves along the surface of the earth. These waves are known as Seismic waves.
15. The process of meeting of negative and positive charges to release huge amount of energy is called electric discharge.
16. An earthquake is a natural phenomenon that occurs because of disturbance deep inside the earth's crust. An earthquake is a sudden shaking or trembling of the earth which lasts for a very short time.
17. The boundaries of the tectonic plates are at high risk of earthquake. These boundaries are called seismic zones or fault zones.
18. (i) Kashmir
(ii) Rajasthan
(iii) Gujarat
(iv) Himachal Pradesh
(v) Uttarakhand
19. When amber is rubbed with fur, it attracts light objects such as hair. In the same way woollen and polyester clothes also attract light objects.
20. Earthquake can cause floods, landslides and tsunamis.
21. Loss of memory, loss of sight or hearing, broken bones etc.
22. An American Scientist Benjamin Franklin established that sparks produced by amber are similar to those produced in sky during a thunderstorm. He flew a kite tied with a silken thread, which had an iron key attached to it on a rainy day. During lightning he felt the shock through iron key, which proved that clouds also carried charges.
23. Seismograph is a device used to measure seismic waves generated by earthquakes. .
24. Tsunami is a giant waves caused by earthquake under the ocean.
25. Earthquake.
26. If we touch the strip standing on the ground (not on any insulating base), then the charge on the strip will pass through our hand to the ground and the strips will contract.
27. The lightning conductor will not work if left hanging in the air because in order to work, it must be connected to the ground by using any metal (copper) plate so that the charge can pass through it to the ground.
28. During lightning, an electric charge may discharge through telephone cords of landline telephone and may become dangerous. Therefore, it is safer to use mobile phones and wireless/cordless phones during lightning.
29. The charged plastic straw will attract another uncharged plastic straw when brought near to it because charges of a different kind attract each other.

30. No, there will not be any repulsion between the aluminum strips. The charged body will not transfer any type of charge to the ebonite rod as the ebonite rod is an insulator. Therefore, there will be no charge on the aluminum strips and no repulsion of the strips will occur.
31. When a plastic refill is rubbed with polythene, it acquires a small electric charge. Similarly, when a plastic comb is rubbed with dry hair, it also acquires a small charge. These objects are called charged objects. In this process of charging the refill and the plastic comb, polythene and hair also get charged.
32. The negative and positive charges meet, producing streaks of bright light and sound. This process is called electrical discharge, the process of electric discharge can occur between two or more clouds or between clouds and earth.
33. The lightning conductor does not allow the charge to accumulate on a building and transfers all the charges to the earth, protecting the building from being struck by lightning.
34. Earthquakes are caused by the movement of plates. The regions which fall on the boundaries of these plates are called danger zones. Earthquakes are most likely to happen in these danger zones.
35. There are two types of charges:
- (i) Positive charge and
 - (ii) Negative charge.
- Charge acquired by a glass rod rubbed with silk is called positive charge and the charge acquired by silk cloth is called negative charge.
36. An earthquake is a sudden shaking or trembling of earth which lasts for a very short time. It is caused by a disturbance inside the earth's crust. There can be a great loss to life and property. Vibrations caused by an earthquake travel in the form of waves along the surface of the earth. These waves are known as Seismic waves. Earthquakes of high magnitudes (7.0 and above on Richter Scale) can cause serious damage to human life and property. Richter scale is used to measure the magnitude of an earthquake. Charles Francis Richter, an American seismologist and physicist is known as the creator of the Richter magnitude scale. It is not possible to predict the occurrence of an earthquake.
37. Lightning (Do you remember, lightning is an electrical discharge ?) can strike telephone cords, electrical wires and metal pipes. During a thunderstorm contact with these should be avoided. It is safe to use mobile phones and cordless phones. However, it is not wise to call up a person who is receiving your phone through a wired phone. Bathing should be avoided during thunderstorm to avoid contact with running water. Electrical appliances like computer, TV, etc. should be unplugged. Electrical lights remaining on do not cause any harm. Air conditioners should be switched off to protect the machinery from damage.
38. It happens because the television screen or computer monitors are charged and have static charges accumulated on them and when we touch them with our hands which are uncharged, the charges transfer through our hands and we get a slight shock.
39. If a positively charged object is brought in contact with the metal clip of an electroscope, the negative charge given earlier to the clip will be neutralised and the diverged strips will contract.
40. Causes of sparking:
- (i) On electric pole wires become loose.
 - (ii) Blowing of wind and shaking the wires.
 - (iii) Looseness of plug in its socket.
41. Lightning conductor is a device used to protect building from the effect of lightning. A metallic rod taller than the building is installed in the walls of the building during construction. One end of the rod is kept out in the air and the other is buried deep the ground. The rod provides easy route for transfer of electric charge to the ground.



42. Interestingly, earthquakes can bring about changes on the earth's surface which are not destructive. New islands, coral beaches and freshwater springs are known to have emerged due to the upheaval caused by the earthquakes.

43. The charging of an object by rubbing it with another object is called charging by friction.

The charging of a glass rod by rubbing it with a silk cloth is an example of charging by friction.

44. A lightning victim carries no electrical charge and cannot electrocute you.

The death by lightning occurs due to cardiac arrest and stopped breathing. You can give first aid by performing CPR and mouth to mouth breathing (resuscitation), which is to be learned from a trained person.

Lightning can cause damage to nervous system, broken bones, burn injuries, loss of hearing or sight, short-term memory loss.

45. When both the straws are rubbed with a dry cotton cloth, they will acquire the similar charge and hence they will repel each other.

46. The areas fall between the boundaries of two plates are called weak zones or seismic or fault zones. In India, the most threatened areas are Kashmir, Western and Central Himalayas, whole of North-East, Rann of Kutch, Rajasthan, Indo-Gangetic plain and some areas of south India.



Fig. 15.8. Movements of Indian earth's plate.

47. Woollen or synthetic sweaters develop static charges (electrons) on them due to friction. While taking off our sweater, these charges move in streams between the sweater and our body, i.e. electric discharge takes place. This results in crackling sound and miniature sparks of lightning.

48. The power of an earthquake is expressed in terms of magnitudes on a scale called Richter scale. Really destructive earthquakes have magnitudes higher than about 7 on the Richter scale. Both Bhuj and Kashmir earthquakes had magnitudes greater than 7.5. Like many other scales in science (decibel is another example), Richter scale is not linear. This means that an earthquake of magnitude 6 does not have twice the destructive energy of an earthquake of magnitude 3. In fact, an increase of 2 in magnitude means 1000 times more destructive energy. For example, an earthquake of magnitude 6 has thousand times more energy than an earthquake of magnitude 4.

49. At the time of thunder negative charges get accumulated near the clouds and positive charges accumulate near ground. When these charges meet electric discharge takes place between ground and clouds. In this process a large amount of energy is released as thunder and lightning.

50. No, there will not be any lightning at all because charge separation cannot take place in conductors. So, all the charges collected at the clouds will continuously transfer through the air, i.e. charge will not accumulate on the cloud and so, there will be no lightning.

51. If aluminum strips are replaced by plastic strips, hence no divergence of the strips will take place, we cannot predict whether the object is charged or not.

52. The electrical charges generated by rubbing produce static electricity. The charges do not move in static electricity while charges move in current electricity.

Static electricity	Current electricity
1. Static electricity is caused by the building up of electrical charges on surfaces of objects.	1. Current electricity is caused by the flow of the electrons along a conductor.
2. Static current happens sporadically and is usually uncontrolled.	2. Current electricity is normally controlled and is used in a wide range of applications.

53. (i) During a thunderstorm contact with telephone cords, electrical wires and metal pipes should be avoided.

(ii) It is safer to use cordless phones and mobiles.

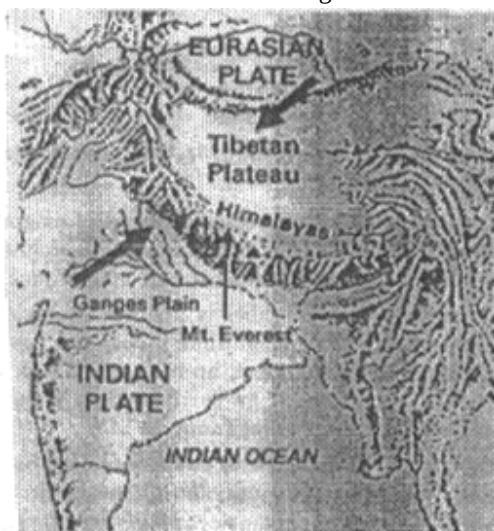
(iii) Bathing should be avoided during thunderstorm.

(iv) Computer, TV etc. should be unplugged.

54. Lightning is a very heavy flow of charge between clouds or between clouds and earth.

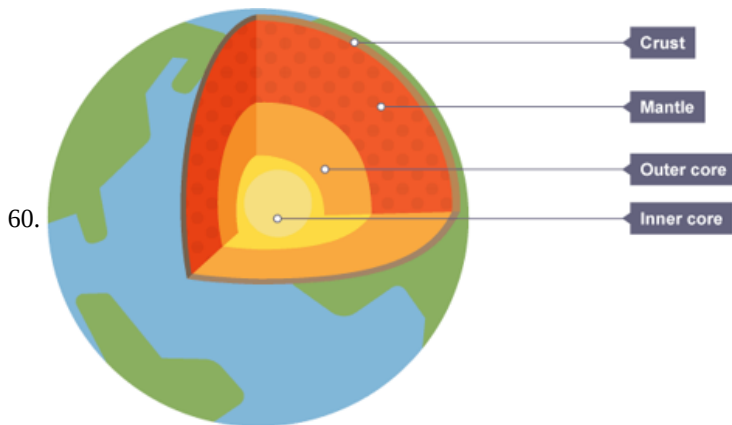
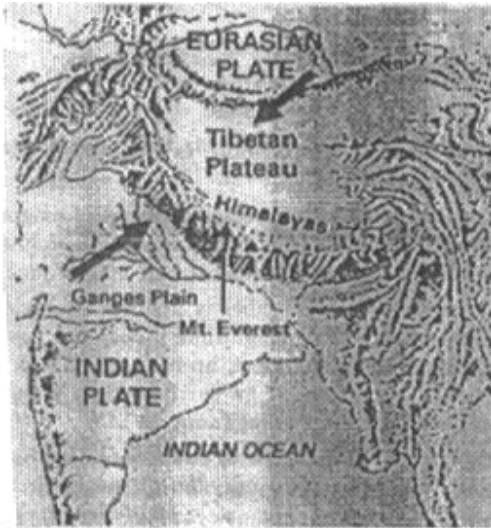
In hilly areas clouds are comparatively closer to the ground than in the plains. Therefore lightning strikes are more frequent in hilly areas.

55. No, scientists have tried lots of different ways of predicting earthquakes but none have been successful so far. They have a pretty good idea of where an earthquake is most likely to hit but they still cannot tell exactly when it will happen.
56. (i) A major earthquake occurred on 26 January 2001 in Bhuj district of Gujarat.
(ii) A major earthquake also occurred in India on 8 October 2005 in Uri and Tangdhar towns of North Kashmir.
57. Some suggestions are:
- (i) Consult qualified architects and structure engineers.
 - (ii) Make the structure symmetrical so that the mass is distributed uniformly.
 - (iii) Use of mud or timber is better than the heavy construction.
 - (iv) The cupboards and shelves are to the walls.
 - (v) Be careful about where to hang wall clocks, photo frames, water heater etc. So that in the event of an earthquake they do not fall on people.
 - (vi) All building should have fire fighting equipment installed.
58. We know that during the development of a thunderstorm there is upward and downward movement of air currents. The vigorous movements of water droplets in the clouds cause separation of charges. By a complex process, not completely understood, positive charges accumulate near ground and negative charges in the clouds.
As soon as negative charges from the clouds and positive charges on the ground meet huge amount of energy is produced as bright light and sound, what we see as lightning. The process is called electric discharge.
The process of electric discharge can occur between two clouds, between clouds and the earth, even between clouds and the human body. Such lightning strokes could be fatal.
59. Since earthquake are caused by the movements of plates, the boundaries of the plates are the weak zones where earthquakes are more likely to occur. The weak zones are also known as seismic or fault zones. In India the areas most threatened are Kashmir, Western and Central Himalayas, the whole of North-East, Rann of Kutch, Rajasthan and the Indo-Gangetic Plane. Some areas of South India also fall in the danger zone.



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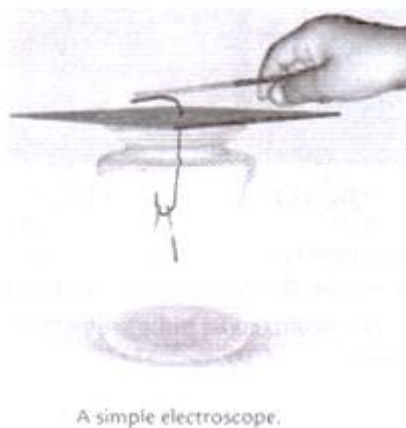
South India also fall in the danger zone.



61. The device which is used to test whether an object is carrying charge or not is called electroscope.

Structure of Electroscope: It is made of a glass jar. A vertical brass rod is inserted into the jar through a cork. The top of the brass rod has a horizontal brass rod or a brass disc. Two gold leaves are suspended from the brass rod; inside the jar.

Working of Electroscope: When the brass disc of the electroscope is touched with a charged object, electric charge gets transferred to the gold leaf through the rod. This results in the gold leaves moving away from each other. This happens because both the leaves have similar charges.

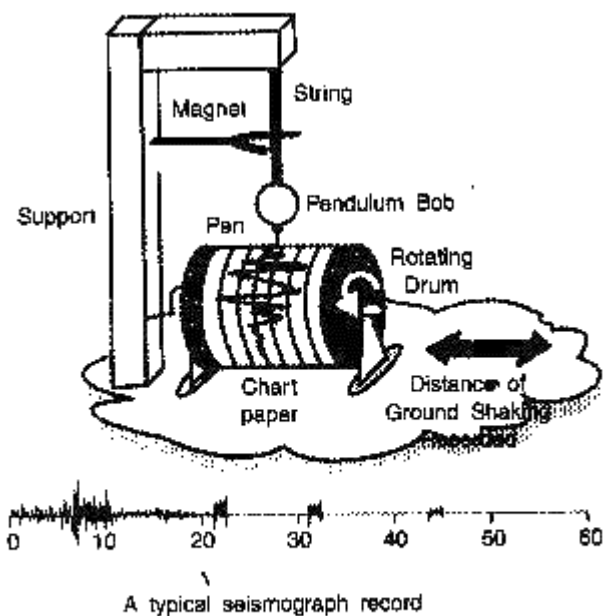


A simple electroscope.

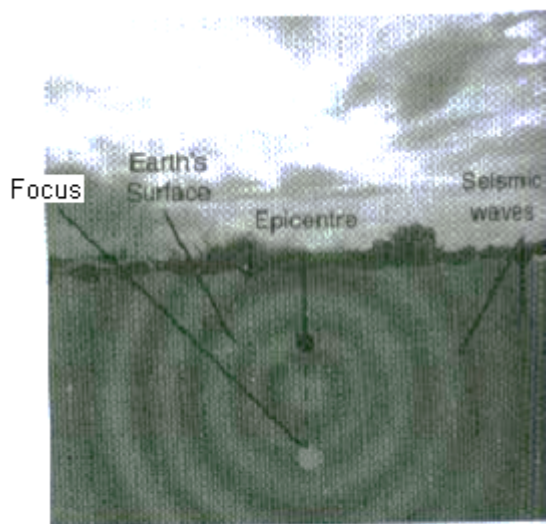
62. The power of an earthquake is expressed in term of a magnitude on a scale called Richter scale.

Yes, it would be recorded by a seismograph. But it is not likely to cause much damage.

The tremors produce waves on the surface of the earth. These are called seismic waves. The waves are recorded by an instrument called the seismograph. The instrument is simply a vibrating rod, or a pendulum, which starts vibrating when tremors occur. A pen is attached to the vibrating system. The pen records the seismic waves on a paper which moves under it. By studying these waves, scientists can construct a complete map of the earthquake, as shown in fig. They can also estimate its power to cause destruction.



A seismograph



Inside the earth

Map of the earthquake

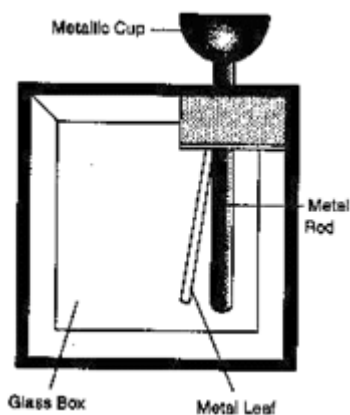
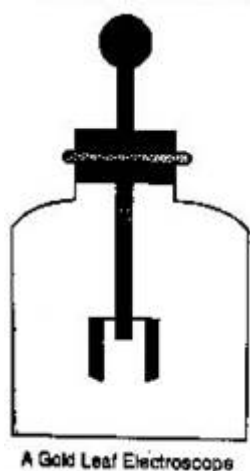
63.
 - No open place is safe during lightning and hence a home or any other building is the safest place during lightning.
 - The first sound of thunder is like a warning call for lightning. Hence, one should rush to a building soon after hearing a sound of thunder. One should come out of home only a few minutes after the last sound of thunder.
 - You should not carry an umbrella (with a metallic handle) during thunderstorm. Metallic handle can be a potential target for lightning.
 - Tall trees and other tall structures are more prone to be hit by lightning. If you are caught in the open, try to hide under a short tree.
 - If you are in open and have no place to hide, then crouch on the ground and keep your head between your knees and hands.
 - Don't take shower during lightning.
 - Use of phone should be avoided during lightning.
 - Television should be disconnected from the antenna.
 - Avoid operating the electric switches during lightning.
64. The instrument used to record the seismic waves is called seismograph. The tremors produce waves on the surface. The instrument is simply a vibrating rod or a pendulum. Which starts vibrating when tremors occur. A pen is attached to vibrating system. The pen records the seismic waves on a paper which moves under it. By studying these waves scientists construct a map of earthquake. They can also estimate its power to cause destruction.

65. Seismograph is a device which records the seismic activities. It is composed of an oscillator, writing device and a roll of paper. The writing device is attached to the oscillator. In case of an earthquake, the oscillator begins to oscillate. This creates oscillation in the writing device; which plots wave-like patterns on the paper. The wave-like pattern is then analysed by seismologist to interpret the various features of an earthquake.
66. Lightning tends to strike the taller objects in an area, such as tall buildings and trees. It's not that those objects attract lightning, it's just that lightning is the result of two narrow channels of opposite charge making contact with each other and the charge flows easier through these objects as opposed to air, so taller the object, lesser air charge has to build through.
67. **Electroscope:** Electroscope is a device used to detect the presence of charge on an object.

Principle: It works on the principle that like charges repel while unlike charges attract each other.

Structure and Working :

An electroscope has a metal rod with thin metal strip or leaf attached to it at the bottom. At the top, the rod enters in a cup. The bottom part of the rod and leaf are enclosed in a glass box for protection. When the knob of the electroscope is touched with a charged ebonite or glass rod, the leaves open out or diverge. Extent of divergence depends upon the amount of charge on the electroscope.



68. Some safety measures are:
- i. We should run to take shelter in the house.
 - ii. We should remain in the covered area.
 - iii. We should not sit in open, on scooters or bike etc.
 - iv. We should take shelter under a small tree while in open.
 - v. If there is no tree or other shelter we should sit with head folded.
 - vi. We should plug out all the electrical appliances during lightning.
 - vii. We should not use wired telephones during lightning. Mobiles and cordless phones are safe.



Safe position during lightning.

69. Precautions to be taken, while we are outside the house during tightening

- i. Do not stand underneath a natural lightning rod such as a pole or a tall tree.
- ii. Stand away from any fountain or any other water body.
- iii. Stay away from tractors or other metal equipment like wire fences, metal pipes, rails, etc.

70. Lightning will not strike the building because charge separation cannot take place in conductors and so, all the lightning falling on the building will reside on the surface of the building. So, there is no need of installing any lightning conductor because all the work of lightning conductor is done by the conducting material itself.

71. During the development of a thunderstorm, the air currents move upward while the water drops move downward. These movements cause separation of charges. The positive charges collect near the upper edges of the clouds and negative charges accumulate near the lower edges. There is accumulation of positive charges near the ground also. When the amount of accumulated charges becomes very large, the air which is normally a poor conductor of electricity, is no longer able to resist their flow. Negative and positive charges meet, producing streaks of bright light and sound. This process is called an electric discharge. The process of electric discharge can occur between two or more clouds or between clouds and the earth. In this way thunderstorm is caused.

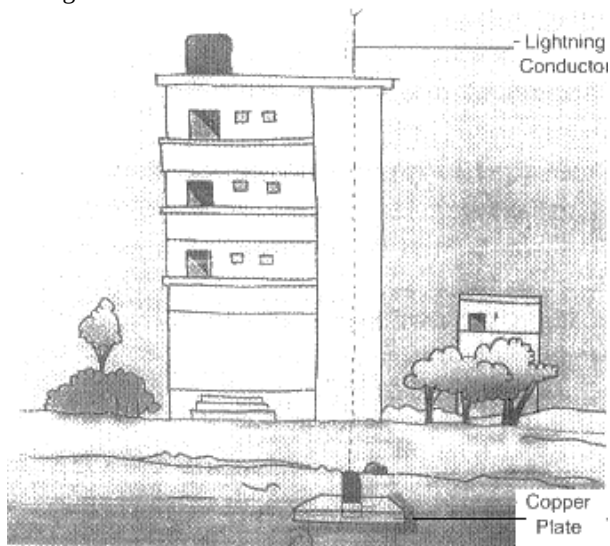


Accumulation of charges leading to lightning.

72. Lightning rod is a device used to secure tall buildings from the effect of lightning. A metallic rod taller than the height of the building to be protected is installed in the walls of the building during its construction. One end of the rod is kept out in the air and the other is fixed in the ground. This provides easy route for transfer of electric discharge to the ground. Several such rods would be required to safeguard a building with a large area.

The metal columns used during construction and wiring and plumbing also give protection to an extent. But do not touch them

during a thunderstorm.



Lightning conductor

73. 1. Take a ball pen refill and rub it against a woolen cloth.
2. Bring this refill near bits of paper.

It is observed that the refill attracts the bits of paper. This activity shows that a charged object attracts uncharged object.

74. In the event that an earthquake does strike, take the following steps to protect ourself.

1. If we are at Home

- a) Take shelter under a table and stay there till shaking stops.
- b) Stay away from windows.
- c) Stay away from tall and heavy objects that may fall on you.
- d) If we are in bed, do not get up. Protect our head with a pillow.

2. If we are Outdoors.

- a) Find a clear spot, away from buildings, trees and overhead power lines. Drop to the ground.
- b) If we are in car or a bus, do not come out. Ask the driver to drive slowly to a clear spot. Do not come out till the tremors stop.

75. The tremors produce waves on the surface of the earth. These are called seismic waves. The waves are recorded by an instrument called the seismograph. The instrument is simply a vibrating rod, or a pendulum, which starts vibrating when tremors occur. A pen is attached to the vibrating system. The pen records the seismic waves on a paper which moves under it. By studying these waves, scientists can construct a complete map of the earthquake, as shown in fig. They can also estimate the energy associated with the earthquake, which determines its power to cause destruction.

